

WASP-39b

R-1810

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_250603 · created 2026-08-03T03:21:50+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460829.7801 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.013
Transit depth (Rp/Rs)^2	3.66 +/- 0.51 [%]
Orbital Inclination (inc)	86.18 +/- 0.63
Airmass coefficient 1 (a1)	1.078 +/- 0.028
Airmass coefficient 2 (a2)	-0.037 +/- 0.016
Scatter in the residuals of the lightcurve fit is	2.65 %
Best Comparison Star	#3 - [303, 445]
Optimal Aperture	2.9
Optimal Annulus	6.89
Transit Duration (day)	0.097 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.191 $\pm$ 0.013 vs 0.1457 (deviation 3.1 σ)
Duration fit vs published	0.097 d vs 0.12 d (deviation 1.86 σ)
Mid-time O–C	23.5 min
Depth SNR	13.1
Residual scatter	2.65 %

Light curve

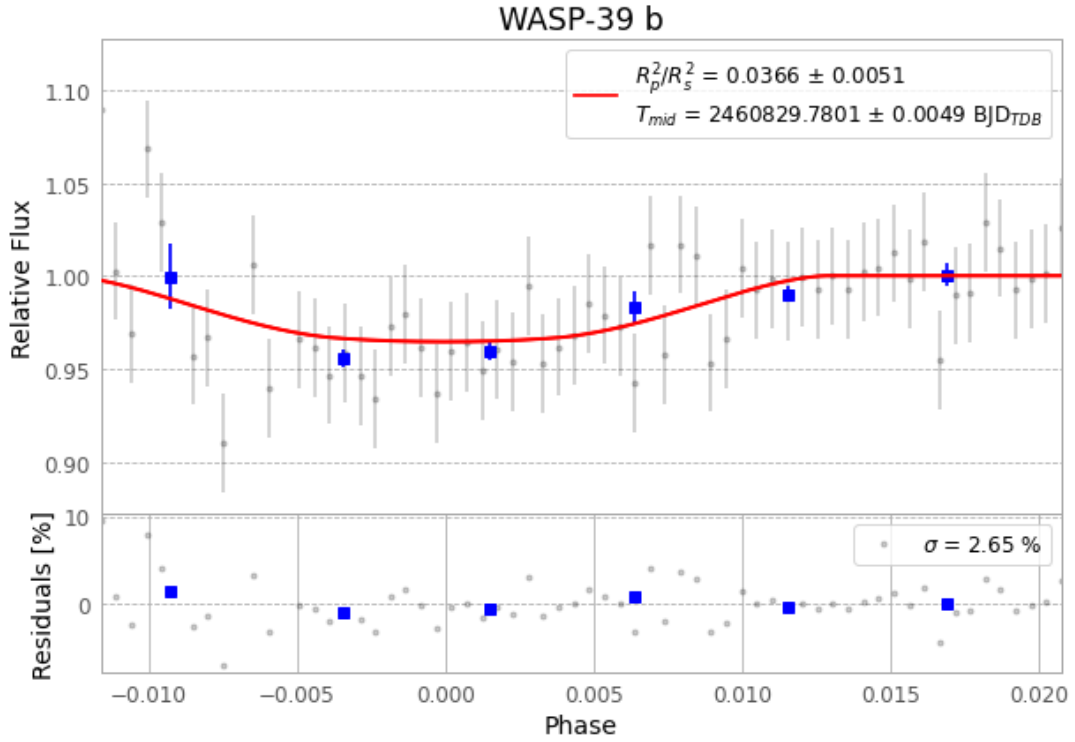


Figure 1 — FinalLightCurve_WASP-39 b_2025-06-02.png (SHA-256 118f0e0205d9e452...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 67eea50ebb05410b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-39250603052715.FITS → WASP-39250603083615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-250603.log (cc88a329a9cf...), FinalLightCurve_WASP-39 b_2025-06-02.png (118f0e0205d9...), FinalLightCurve_WASP-39 b_2025-06-02.csv (8886026452fb...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 03:21Z.